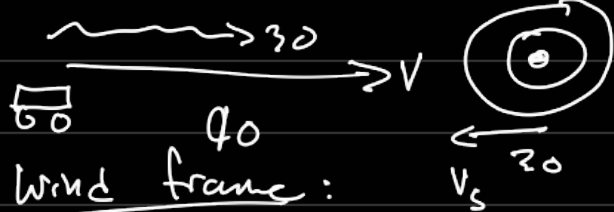


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* Wind frame:
car: 60 km/s
siren: -30 km/s

$$500 \text{ Hz} = f_0$$

$$f_{\text{new}} = \left(\frac{c_s + v}{c_s - |v_s|} \right) f_0$$

$$c_s: 340 \text{ m/s}$$

$$v: 60 \text{ km/s}$$

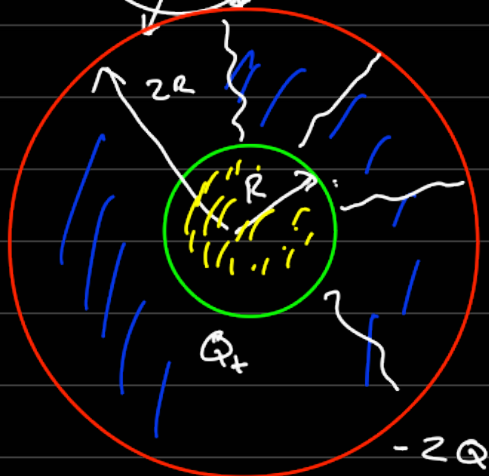
$$|v_s| = 30$$

① Gauss' Law: $\Phi_{\text{flux}} = \sum \vec{E}_i \cdot \vec{A} = \frac{Q_{\text{enclosed}}}{\epsilon_0}$



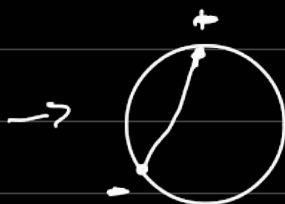
$$E 4\pi R^2 = \frac{-e}{\epsilon_0} \Rightarrow E = \frac{-e}{4\pi \epsilon_0 R^2}$$

(a) $E: |x| < R$

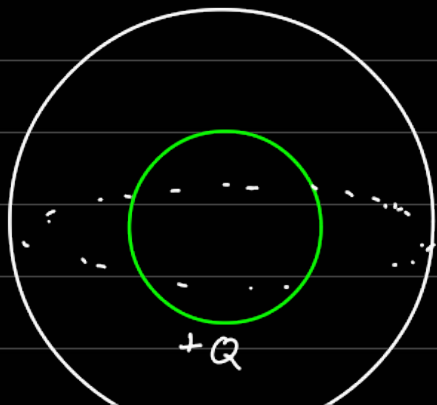


$$\Phi_{\text{flux}} = E \cdot A = Q_{\text{enc}} = 0$$

$$E_{\text{in}} = 0$$



② $R < |x| < 2R$

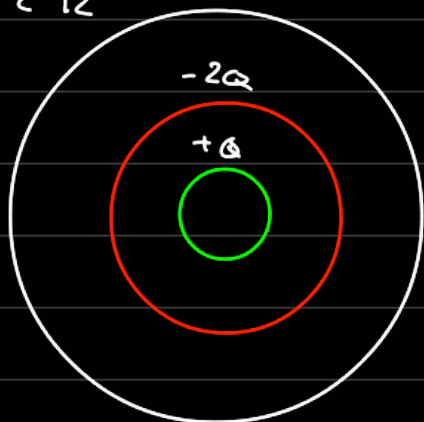


$$EA = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$E_{\text{in}} = \frac{Q}{4\pi \epsilon_0 |x|^2}$$

(C) $|x| > 2R$

$$E_{\text{enc}} 4\pi |x|^2 = \frac{Q_{\text{enc}}}{\epsilon_0}$$



$$Q_{\text{enc}} = +Q + (-2Q)$$

$$Q_{\text{enc}} = -Q$$

$$E_{\text{enc}} = \frac{-Q}{4\pi\epsilon_0 |x|^2}$$

(ii) Potential: $\vec{F} = -\nabla\phi$

$$F = -\frac{d}{dx}\phi$$

$$* F = (-Q) E = -\frac{d}{dx}\phi *$$

$$E \sim \frac{1}{|x|^2} = -\frac{d}{dx} \frac{1}{|x|}$$

$$-Q \left(\frac{\tilde{Q}}{4\pi\epsilon_0} \left(-\frac{d}{dx} \frac{1}{|x|} \right) \right) = \left(-\frac{d}{dx} \phi \right)$$

$$\phi = \frac{-Q \tilde{Q}}{4\pi\epsilon_0 |x|}$$

$$\phi|_{-\infty}^0 = \phi|_{-\infty}^{2R} + \phi|_{2R}^R + \phi|_R^0 \rightarrow Q=0$$

$$= \cancel{\phi(2R)} - \phi(\infty) + \phi(R) - \cancel{\phi(2R)}$$

$$+ \cancel{\phi(0)} - \cancel{\phi(2)}$$

$$= \phi(R) - \phi(\infty)$$

$$[\phi(\infty) = \frac{-Q \tilde{Q}}{4\pi\epsilon_0(\infty)} \rightarrow 0]$$

$$= \phi(R) = \frac{(-Q)(Q)}{4\pi\epsilon_0 R} = \frac{-Q^2}{4\pi\epsilon_0 R}$$

